

Coursework Report

Topic: B2B Technology Sales, Sales Development

1. Business Perspective

This coursework examines the way of how data science can support the decision-making process in a real business case. This report explores a business-to-business (B2B) technology sales environment, with a focus on the role of sales development representatives (SDRs). SDRs face the daily challenge of qualifying fast and prioritising a large number of potential leads/customers within a short time. This is often executed with incomplete information and limited sources. From my experience of working in B2B technology sales environment, I have observed that SDRs usually rely on intuition and personal judgment. Thereby, using inconsistent prioritisation methods that result in inefficiencies, missed opportunities, and unpredictable sales performances. Therefore, this project consists of how applying data science principles can help SDRs to make more consistent, evidence-based decisions, ultimately improving lead quality, qualification speed, and conversion rates.

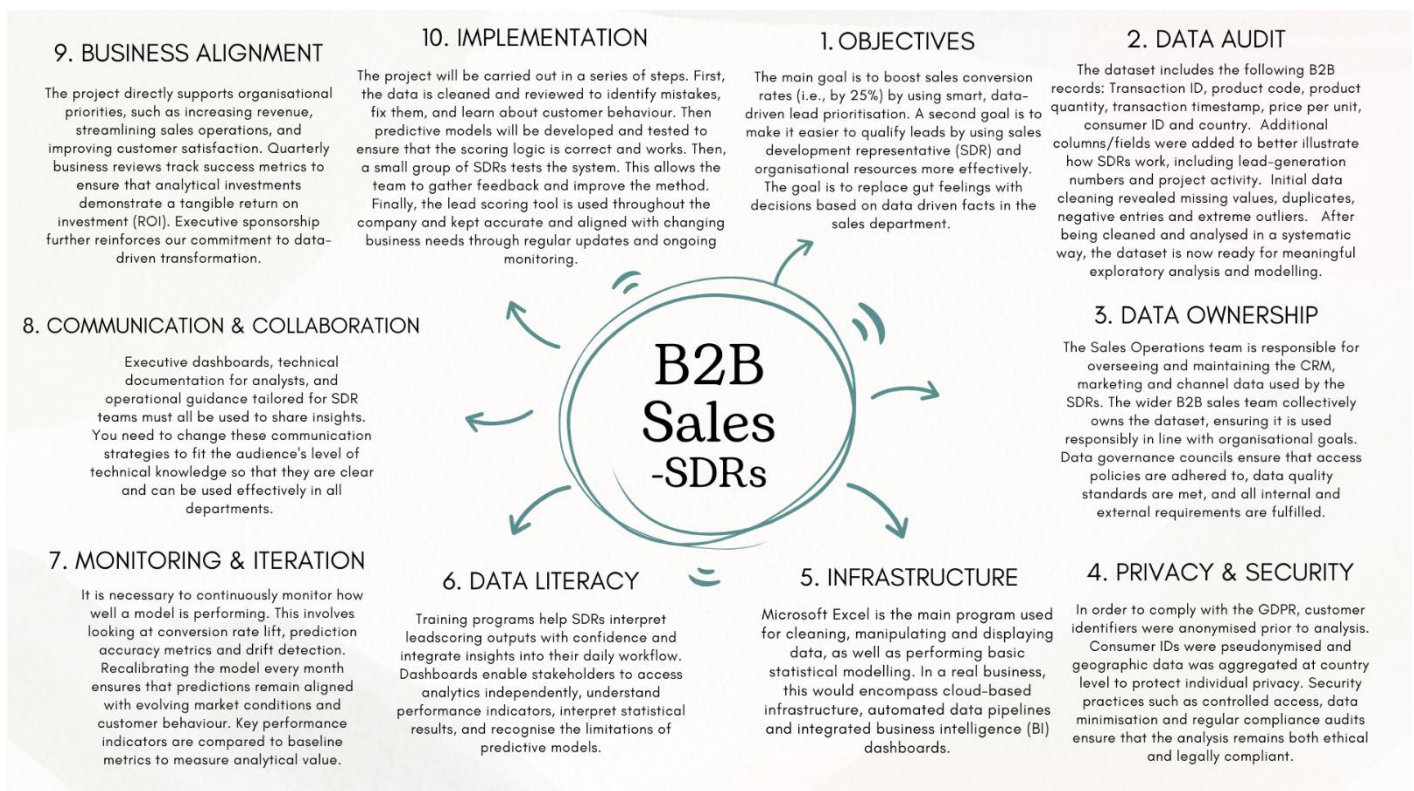


Image 1 – Business Perspective B2B Sales

Image 1 illustrates several steps of the entire data science lifecycle that should be used to solve these business problems. The main goal is to make lead conversion more accurate and efficient by prioritising and using data to find useful information. Another key aim is to support SDRs (Sales Development Representatives) in identifying which accounts to prioritise and understanding the reasons behind those decisions. This enables them to reduce the time spent on lead qualification and enhance overall conversion rates.

Additionally, the dataset was improved and cleaned by identifying and addressing issues such as missing values, negative quantities and duplicates, as well as by locating and removing outliers. Addressing and removing these errors ensures the reliability of the dataset and subsequent analyses. Therefore, data ownership is crucial for ensuring the quality and accessibility of the dataset, which leads to better results and supports SDRs in reaching their goals.

The data has also been anonymised to maintain privacy and comply with GDPR principles. To align with real-world practices, the quality variables have been pseudonymised and aggregated to minimise reidentification while protecting customer privacy.

This report provides an infrastructure based on Microsoft Excel and its internal tools. It is a practical environment for cleaning, transforming, managing and analysing datasets. To use, understand and implement the data all relevant personas (SDRs, managers and stakeholders) require training and enablement. This shows the potential and limitations of the predictions.

Additionally, it involves a monitoring and iteration stage that highlights the importance of frequently monitoring the model and deploying it in a real business environment. Regular (i.e. monthly) monitoring of the model's accuracy is essential for improvement and eliminating mistakes. This will lead to better predictions and help the sales team improve lead conversion and efficiency. Also, illustrations will help stakeholders to correctly interpret the data.

In summary, to achieve the best possible outcome from the implementation of data science, it is crucial to follow a plan involving preparing and cleaning the data, building productive models, piloting the lead scoring system, and implementing it within the organisation. To improve the efficiency of SDRs and making the sales pipeline more predictable is directly aligned with the business's revenue goal.

2. Data Reprocessing

This step demonstrates the process of re-processing the dataset, which should reflect transactional behaviour in a B2B sales environment. I therefore maintained and followed the preprocessing structure of the dataset, which involved cleaning, transforming and handling incorrect values in Excel. Each step is documented on a separate Excel sheet to enable a full understanding of the dataset transformation.

I started implementing the raw dataset into the 'RAW_DATA' sheet so that I could access it at any time. I left it unchanged on purpose, to use as a reference point throughout the project. I used filters and Excel functions like the conditional formatting > Go To Special > Blanks method to find missing values, inconsistencies, and irregularities, and then highlight those missing values. Incorrect values can alter averages and make subsequent analyses less accurate. To fill in the missing values and keep the dataset accurate, I then found the average of the "ProductQuantity" and "PricePerUnit" variables. I also used a different method for unique identifiers like "ConsumerID" by putting "N/A" in for missing values. These values don't change the results of the analyses, but they are important for keeping the dataset's integrity. This lets the row stay in the dataset while also showing that it is missing information. I also got rid of duplicate values so that the number of transactions wouldn't go up. The "Data_CleaningStep1" sheet contains these steps.

Afterwards, I identified the outliers in the dataset. Outliers are values that are unusually high or low and represent data entry errors, so they must be replaced or removed. I decided to remove the outliers to maintain the accuracy of the dataset. I used the interquartile range (IQR) method to calculate Q1 and Q3, and determined the outlier thresholds using the formula $Q3 + 1.5 * IQR$. The conditional formatting technique highlighted the outliers using colours. These values have been removed to ensure the dataset is clean and to avoid distorting the descriptive statistics and predictive modelling. You can see this step in the 'Clean_Data' sheet, where all outliers have been removed.

To transform the cleaned dataset, see 'Transformed_Data' sheet, I applied the normalisation, standardisation and binning techniques. Therefore, I have added those in extra columns and scaled the normalisation of the variables to the range [0,1] using the formula $= (x - \min) / (\max - \min)$. This is useful if the variable is on a different scale. Whereby, the standardisation technique converts values into z-scores that show the quantity of standard deviations in each value from the mean. This is an essential step for analytical methods, such as PCA, to prevent the domination of large-scale variables. The binning technique, on the other hand, converts continuous variables into categories of 'HIGH' and 'LOW'. This makes it more visually appealing to non-technical stakeholders and helps implement rule-based models or segmentations.

Transactional fields cannot fully represent the customer engagement or lead quality of a dataset. Therefore, I added two new real-world data of 'Leads_Generated' and 'Total_Projects' that help to reflect a real-life scenario of buying intends of prospects and/or customers for SDRs and sales teams. 'Leads_Generated' illustrates how many leads have been qualified throughout the purchasing process and associates the the accounts and likelihood of conversion rates. Whereby, 'Total_Projects' shows the extent to which the

customer has already engaged with the company. This helps sales teams to identify high-value accounts. Those features are widely adopted across companies in the CRM and are used with frameworks such as the BANT qualifications to qualify and convert leads (scoring models).

To maintain realism, I used samples from 15 rows of two new columns from real-life datasets on Kaggle; see (Ezo, 2025) & (Miltos Pontikis, 2025). As those samples are too small for the already existing dataset, I used bootstrapping to extend those. Additionally, in the sheets of 'Bootstrapping', 'Bootstrap_Sample1' and 'Bootstrap_Sample2', you can see that I used formulas such as `=INDEX(sample_range,RANDBETWEEN(1,n))` in order to generate synthetic datasets for each new column to match the length of 127 rows. I also changed the bootstrapped version in the "New_CleanData" sheet. This made sure that the changed dataset had realistic, varied engagement metrics that could be used for further analysis.

Lastly, I performed a dimensional reduction using the Principal Component Analysis (PCA) to explore/discover if the modified dataset would be simplified without losing important information. Due to a non-built-in PCA tool, I carried those steps out manually. First, I standardised numerical variables using the z-scores to make sure they are on the same scale. Then, I used the Covariance tool to understand how the variables move together. Whereby, the use of Excel's matrix functions helps to discover the eigenvalues and eigenvectors that form the basis of the principal components. Therefore, the analysis illustrates that the first principal component explains approximately 64,6% of the total variance, while the second only 35,4%. Both variables are substantial, non-overlapping, and reducing the dataset to a single dimension would lead to meaningful information loss. This confirms that the chosen variables reflect different aspects of the customer behaviour, and it makes sense to remain separate in later modelling.

3. Data Visualisation

The Data Visualisation part is very important for making the cleaned and enhanced dataset easier to understand. See the "Data_Visualisation" sheet for more information. These charts show patterns in the data and make it easier for SDRs, sales teams, and BI/analysts to share their findings with other people.

First, I created a histogram of 'ProductQuantity'. Histograms are helpful charts to visualise frequency distributions due to their group values into intervals/bins and illustrate the occurred frequency. This histogram reveals a right-skewed distribution that shows that most orders are between 1 and 14 units with only a few larger orders up to 48 units. This is a typical pattern in B2B sales, where small, frequent orders are common and bulk purchases are less frequent. SDRs can use this information to find patterns that big orders might show and important accounts that need to be followed up on right away.

In addition, the created bar chart compares the number of transactions in different countries. Bar charts are helpful for clear comparisons between categories and support identifying which region contributes to the most purchases in the dataset. You can see that the UK dominates with over 90% of the transactions, while countries such as Germany, Switzerland, Sweden and France contribute minimally. This indicates that SDRs should adopt different pipeline strategies in different regions. Managers could also use this information to identify dependency risks and explore opportunities in underrepresented markets.

I also added a pie chart showing each country's percentage share. Pie charts are a practical illustration for 'part-to-whole relationships. This one also clearly shows the UK dominance. Looking at both bar and pie chart, it ensures the stakeholders with different visual preferences to understand and interpret the information.

The scatter plot shows the relationship between price per unit and product quantity. It is helpful to understand the relationship between two variables. The trendline showed a weak negative correlation, which meant that higher prices were linked to slightly lower amounts, but the link wasn't very strong. This shows that price alone is not a reliable sign of how many orders or purchases there will be. This information is helpful for SDRs and sales managers because it shows that only looking at price when deciding which leads to follow up on is not enough. It shows that price isn't the only thing that matters when deciding which leads to follow up on; engagement-related factors are more important.

Finally, I also made a residual plot to see if the regression model was correct. The residuals were spread out fairly evenly, which means that there were no major violations of the linearity assumption. However, the model's predictive power is limited is confirmed by the R^2 variable/value. These data visualisations provide a clear and intuitive understanding of the dataset together. It also supports the interpretation of the statistical modelling results.

4. Statistical Modelling

This part involves the statistical modelling that includes the construct of confidence intervals, hypothesis testing, performing a correlation analysis and creating a regression model. Those analyses support to identify the quantity of the characteristics of the dataset and evaluates the accuracy of certain variables predicting others.

First, I figured out the confidence interval variables for the dataset. The mean for ProductQuantity was 11.28 units, and the standard deviation was 11.01. This gave a wide 95% confidence interval of [9.38, 13.18]. The histogram shows that the number of orders can change a lot, which is why we can see a wide range here. The confidence interval for PricePerUnit, on the other hand, was much narrower, at [0.94, 1.05]. This means that the prices in the dataset were relatively stable.

Afterwards, I conducted a one-tailed hypothesis test to determine whether the average order quantity exceeded the business-relevant benchmark of 10 units. This threshold shows the smallest order size that businesses think is worth their time and money. The results of the z-score were 1.31, with a p-value of 0.095. This exceeds the significance level ($\alpha = 0.05$). Therefore, I could not reject the null hypothesis. While the sample mean was slightly above the benchmark, there was insufficient statistical evidence for the conclusion of a significant difference. This finding shows that SDRs cannot assume that customers usually place high-volume orders. Instead, they should focus on broader engagement metrics and relationship building when evaluating leads.

I also used the =CORREL function to look at how the variables are related and figure out the correlations. There was a weak negative correlation of about -0.292 between PricePerUnit and ProductQuantity. The data indicated a negligible correlation between TransactionTimestamp and the other variables. This suggests that order quantities do not show a clear pattern based on the time. It appears that purchasing behaviour is more likely to be driven by non-price factors, such as customer engagement, the number of active projects, and customer relationship maturity.

I also created linear regression model with "PricePerUnit" as the independent variable and "ProductQuantity" as the dependent variable. The regression equation was: $\text{ProductQuantity} = 20.96 - 9.76 \times \text{PricePerUnit}$. While the coefficient was statistically significant ($p \approx 0.0009$), the R^2 value was only 0.085. This means that the dataset model only accounts for 8.5% of the differences in order quantity. This confirms our hypothesis that price alone is insufficient to predict customer behaviour, and therefore should not serve as the primary criterion for SDR prioritisation. Therefore, it would be useful to use a model that has more information variables such as industry segment or channel source. Even though these extra variables are not used in the regression modelling for this coursework, they set the stage for a better lead-scoring model in future projects.

Summarised, the statistical modelling confirms the initial findings that transactional/numeric values/variables itself are not enough to fully explain customer/prprospects behaviour in a B2B technology environment. To be more reliable and actionable, effective lead-scoring systems require richer datasets and multi-variable models.

5. Conclusion

Concluding, this paper demonstrates how core data science principles are applied to a real-life scenario, in relation to a B2B technology sales scenario, to help SDRs/Sales Development team to prioritize leads more efficiently and effectively. I started with a raw dataset and systematically cleaned it by filling in missing values, getting rid of duplicates, and finding and fixing outliers. Then I normalised, standardised, and put the data into bins. Next, we added two business-relevant variables, "Leads_Generated" and "TotalProjects," to the dataset to make it more useful. The next step was to use bootstrapping to expand the new features and integrate them into the modified dataset, with the aim of making it more suitable for analysis.

To provide valuable insight through data visualisations how things are distributed geographically, how orders are placed and how these things are connected. But the numbers also showed that you can not use transactional variables to guess how customers will act. The regression model also showed that price alone is not a good predictor. This shows once again how important it is to use multi-variable models in real-world lead-scoring systems.

This project shows that how data-driven insights can improve the consistency of lead qualification, reducing reliance on SDRs intuition and enhancing the day-to-day decision-making process. This project aligns with strategic goals of businesses like improving conversion rates, increasing the operational efficiency and develop a more predictable sales pipeline.

The project shows that in order to move from relying on intuition to using data to decide what features to prioritise, that a clean dataset and its features can deliver meaningful insights through multi-variable models. Nevertheless, organisations can begin to develop transparent, interpretable, evidence-based decision-making frameworks using basic tools such as Excel, thereby greatly improving B2B sales performance.

6. References

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